

# PARKFNB

## Competitive & operations research

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Yeh document do cheezein cover karta hai: (1) competitors kya kar rahe hain aur usse humein kya badalna chahiye, (2) asli production systems booking ko physical access se kaise jodte hain. Har number ke source aakhir mein diye hain.

### TEEN SABSE ZAROORI NATEEJE

1. Deck ka market-size number baaki har research firm se  $\sim 17\times$  zyada hai — sabse urgent fix.
2. Direct competitor Parkobot ₹10/hr par hai; hamara deck ₹40/hr maanta hai.
3. Commission zaroori nahi — Rapido ne apne do segments par use chhod diya hai.

### 01 · SABSE URGENT

## Market-size ka number verify nahi hoga

Deck kehta hai **\$4.9B (2025) → \$12.7B (2031) @ 17.6% CAGR**, aur source likha hai "MobilityForesights, IMARC Group, Astute Analytica". Teeno check kiye:

| RESEARCH FIRM      | INDIA SMART PARKING MARKET | CAGR          | DECK SE FARQ             |
|--------------------|----------------------------|---------------|--------------------------|
| MobilityForesights | \$4.9B → \$12.7B           | 17.6%         | deck yahi cite karta hai |
| IMARC Group        | \$289M → \$883M            | 13.2%         | $\sim 17\times$ chhota   |
| MarketsandMarkets  | \$284M → \$566M            | 14.8%         | $\sim 17\times$ chhota   |
| 6Wresearch         | —                          | $\sim 14.3\%$ | CAGR bhi kam             |

### PROBLEM SIRF BADA NUMBER NAHI HAI

Deck IMARC ko supporting source bata raha hai — jabki IMARC ka apna number hamare claim ko  $17\times$  se kaatta hai. Koi bhi investor jo dono naam Google karega, 10 minute mein yeh contradiction dhoondh lega. Phir woh baaki har number par shaq karega.

### Do raaste:

- A — Conservative base lo: \$284–289M (2025). Chhota lagta hai par defensible hai, growth 13–15% ab bhi hai.
- B — Apna TAM khud banao: "top 20 sheher  $\times$  X lakh cars  $\times$  Y paid-parking events/mahina  $\times$  ₹Z". Investors bottom-up TAM ko third-party report se zyada respect karte hain.

**Salah:** Option B. Hum ek hardware + marketplace company hain, "smart parking systems" market ka hissa nahi — woh report barriers aur sensors ka market bech rahi hai. Report citation footnote mein chhod do.

## Rapido ne commission chhod diya — do segments mein

\$3B valuation (2026 Series F), 2M+ rides/din. Asli seekh revenue model mein hai:

| SEGMENT   | MODEL           | CAPTAIN KYA DETA HAI                     |
|-----------|-----------------|------------------------------------------|
| Bike taxi | Commission      | 15–20% per ride                          |
| Auto      | Zero commission | ₹9–29 / din subscription, fare 100% apna |
| Cab       | Zero commission | ₹500 / mahina, jab kamai ₹10,000+ ho     |

Baaki Rapido se: do alag apps (Rider + Captain) — hamare jaisa hi. Matching 2–3 second mein honi chahiye warna user chhod deta hai. Onboarding = DL + RC + insurance + Aadhaar upload → verify → live. Payment cash + wallet + UPI.

## Duniya aur India mein kaun kya kar raha hai

| COMPANY        | SUPPLY                                     | PAISA KAISE                                      | PAIMANA                                            |
|----------------|--------------------------------------------|--------------------------------------------------|----------------------------------------------------|
| JustPark · UK  | 45,000 private driveways + commercial      | Driver se booking fee AUR owner se commission    | \$10.4M rev (2024), 5.5M users, 86 staff — 2006 se |
| SpotHero · US  | Commercial garages — P2P nahi              | 15–30% commission operators se                   | North America                                      |
| Park+ · India  | Societies                                  | FASTag, insurance, challan, EV — parking se nahi | ₹180.4 Cr rev FY25, \$53M raised                   |
| Get My Parking | Commercial lots                            | White-label app + IoT GateKit + cloud (B2B)      | 250 lots / 10 metros, \$6M raised                  |
| Parkobot       | Institutions + commercial (driveways nahi) | ₹10/hr · ₹2,999/mo for 300 hrs                   | 1,600 spots Kolkata, ₹2.09 Cr raised               |

**DO CHEEZEIN JO HAMARE PLAN KO SEEDHA KAATTI HAIN**

Parkobot ₹10/hour par hai — hamara deck ₹40/hour maanta hai. Agar ₹10 market rate ban gaya to per-bay GMV ₹4,800 se ₹1,200 ho jaata hai, aur poori unit economics dobara likhni padegi.

Aur Parkobot ka supply private driveways nahi hai — Park Street, Passport Office, La Martinière, Diamond Plaza. Yaani institutions, jahan density hai: ek deal = 200 bays.

JustPark ka number bhi dhyan se dekho: 2006 mein shuru, 2024 mein revenue \$10.4M, staff sirf 86. Duniya ka sabse successful P2P driveway marketplace ko \$10M tak pahunchne mein 18 saal lage. Yeh business dheemi chakki hai, rocket nahi — deck ke 18-mahine ke targets iske saath rakh kar dekho.

## Teen cheezein badalni chahiye

| KYA                                         | KYUN                                                                                                                                                                                                                                                               | KAAM KITNA       |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Commission → subscription socho</b>      | Rapido ne chhoda, Parkobot subscription par hai. Owner-side subscription (₹199–499/bay/mahina) mein payment flow, escrow, payout engine, PSP reconciliation — kuch nahi chahiye. Backend mein commission field hai hi nahi, to abhi decide karne ka sahi waqt hai. | Faisla + 1 hafta |
| <b>Pricing range dikhao, ek number nahi</b> | ₹40/hr justify karna padega (better location, guaranteed slot, hardware security) ya economics dobara. Deck mein ₹10–40 by location type + sensitivity dikhao.                                                                                                     | Deck edit        |
| <b>Driveways se pehle institutions</b>      | Parkobot, SpotHero, Get My Parking — teeno commercial/institutional par gaye. Density: ek Passport Office = 200 bays, ek driveway = 1 bay. Hamara Property→Space hierarchy yeh already support karta hai, sirf GTM ka order badlega.                               | GTM, code nahi   |

**Ek positive:** Park+ ka pivot hamare liye raasta hai, warning nahi. Parking se customer aata hai, paisa car services se. Hamare consumer app mein vehicle garage already bana hua hai.

## Ek usool: barrier ke saamne network par bharosa mat karo

Har production system yahi karta hai. Booking hone ke waqt grant phone mein AUR device mein — dono jagah pehle hi pahunch jaata hai. Gate par sirf verify hota hai, poocha nahi jaata.

| SYSTEM               | KYA KARTA HAI                                                                                |
|----------------------|----------------------------------------------------------------------------------------------|
| ANPR parking systems | Booking hote hi authorised plates ki whitelist controller par push hoti hai                  |
| Airbnb + Yale/August | Reservation par code pehle hi generate hokar lock mein jaata hai, checkout par mit jaata hai |
| BookFlowGo           | Queue-based push: "supplier API down ho to booking rukti hai, customer nahi"                 |

## Andar ghusne ke hamesha 3–4 raaste hon

Parkable ke paas chaar hain — app tap, ANPR, QR, visitor link. Ek fail ho to agla chalta hai. Kabhi ek hi tareeka mat rakho: basement mein signal nahi hota, plate par dhool hoti hai, phone ki battery marr jaati hai.

| RAASTA            | KAAM                                             | KAB FAIL                           |
|-------------------|--------------------------------------------------|------------------------------------|
| 1 · ANPR          | Plate padho, khud khulo — driver kuch nahi karta | dhool, angle, raat (~85–93% India) |
| 2 · BLE / app tap | Geofenced button, <1.5 sec, net nahi chahiye     | phone nahi / battery khatam        |
| 3 · QR            | Gate par sticker ya driver dikhaye               | camera kharab                      |
| 4 · PIN code      | SMS mein, keypad ya guard — hamesha chalta hai   | —                                  |

### ANPR KO AKELA FAISLA MAT KARNE DO

Match ho to check-in tez karo. Na match ho to agle raaste par jao — entry deny kabhi mat karo. Ek dhool lagi plate ki wajah se customer gate par phansa = refund + bura review + support ticket.

### Do detail jo research se mili aur pehle miss ho gayi thi:

- App ka open button geofenced ho — "location check so it only works on site". Warna koi ghar baithe barrier khol dega.
- Ops ko remote-open button chahiye (hardware phansega, plate nahi padhega) — par har override log ho, kis ne kiya dikhe, aur reason maange.

## Kaun kya control karega

| KAAM                   | DRIVER   | OWNER  | STAFF  | OPS  | AUTO               |
|------------------------|----------|--------|--------|------|--------------------|
| Space list karna       | —        | haan   | —      | —    | —                  |
| KYC approve            | —        | —      | —      | haan | —                  |
| Price set karna        | —        | haan   | —      | cap  | —                  |
| Booking approve/reject | —        | haan   | —      | log  | instant-book       |
| Barrier kholna         | site par | haan   | haan   | log  | ANPR match         |
| Check-in               | —        | manual | manual | log  | access event se    |
| No-show mark           | —        | haan   | haan   | log  | window + koi entry |

|                         |   |   |   |      |        |
|-------------------------|---|---|---|------|--------|
|                         |   |   |   |      | nahi   |
| <b>Refund</b>           | — | — | — | haan | policy |
| <b>Payout release</b>   | — | — | — | run  | T+48h  |
| <b>Firmware rollout</b> | — | — | — | haan | —      |

**Site staff ek alag credential class hai.** Guard ka access booking se nahi aata — permanent hai, site se bandha hua. Usko sirf check-in, gate kholna aur no-show mark milna chahiye — na pricing, na payout, na listing.

## Device system se kaise judta hai

Ek lock chalana aasan hai. Hazaar chalane ke liye lifecycle chahiye — yeh industry ka standard claim-based fleet provisioning pattern hai:

| STEP                     | KYA HOTA HAI                                                                            |
|--------------------------|-----------------------------------------------------------------------------------------|
| <b>1 · Factory</b>       | Shared claim credential + server ka public key flash karo, QR sticker chipkao           |
| <b>2 · Pehla connect</b> | Claim se unique per-device certificate mile; claim ab bekaar                            |
| <b>3 · Bind</b>          | Installer QR scan kare, device ↔ bay bind ho — yeh insaan ka step hai aur rehna chahiye |
| <b>4 · Live</b>          | Roz ~300 bytes whitelist neeche, ~400 bytes events upar                                 |
| <b>5 · OTA</b>           | Cohort mein bhejo, signed, dual-partition, metrics bigde to turant roko                 |

**DO GALTIIYAAN JO MEHNGI PADTI HAIN**

Firmware mein ek shared secret sabke liye mat rakho — ek unit khula to poora fleet khula. (Hamare config.h mein abhi CMD\_TOKEN waisa hi hai; firmware README ne khud yeh flag kiya hua hai.)

Sab devices ek saath activate mat karo — carrier throttle kar dega. Staggered rollout karo.

Aur: har minute "main zinda hoon" ping mat bhejwao — battery aur data dono jaata hai. Chup rehna hi alarm hai: agar N din se kisi device ne sync nahi kiya to admin panel mein alert.

### 05 · FAILURE MODES

## Kya bigdega aur tab kya hoga

| KYA BIGDA                         | SYSTEM KYA KAREGA                                     | KAUN SAMBHALEGA   |
|-----------------------------------|-------------------------------------------------------|-------------------|
| <b>Basement mein signal nahi</b>  | Kuch nahi — grant cached hai, BLE/QR chalega          | —                 |
| <b>Device offline (N din)</b>     | Alert; us bay par nayi booking rok do                 | Ops → field visit |
| <b>Device battery khatam</b>      | <20% par pehle alert; bay auto-block                  | Ops + owner       |
| <b>ANPR plate nahi padha</b>      | Agle access method par gir jaata hai                  | —                 |
| <b>Phone ki battery marr gayi</b> | PIN code SMS mein — keypad ya guard                   | Site staff        |
| <b>Barrier mein rukawat</b>       | Firmware motor band karta hai, fault event bhejta hai | Ops remote open   |
| <b>Payment gateway down</b>       | Booking awaiting_payment mein queue, 15 min TTL       | —                 |
| <b>Backend down</b>               | Purani bookings chalengi; events device par queue     | Ops               |
| <b>Do gaadi ek bay par</b>        | Unauthorized event → owner alert                      | Owner / staff     |

**Dhyan do:** zyadatar rows mein "kaun sambhalega" khaali hai. Achha operations design wahi hai jahan zyadatar failures kisi insaan tak pahunchte hi nahi.

## 06 · SOURCES

Sab public sources, September 2026. Deck mein daalne se pehle khud verify kar lena — market research numbers jaldi purane ho jaate hain.

### Market size

MobilityForesights — [mobilityforesights.com/product/india-smart-parking-market](https://mobilityforesights.com/product/india-smart-parking-market)

IMARC Group — [imarcgroup.com/india-smart-parking-systems-market](https://imarcgroup.com/india-smart-parking-systems-market)

MarketsandMarkets — [marketsandmarkets.com/Market-Reports/geography/parking-management-market/India](https://marketsandmarkets.com/Market-Reports/geography/parking-management-market/India)

### Rapido

StartupTalky — [startuptalky.com/rapido-business-model](https://startuptalky.com/rapido-business-model)

Rapido Business Model Explained 2026 — [rahulmalodia.com/blog/rapido-business-model](https://rahulmalodia.com/blog/rapido-business-model)

RaftLabs — [raftlabs.com/blog/build-app-like-rapido](https://raftlabs.com/blog/build-app-like-rapido)

### Global parking

JustPark Parking Ltd — UK Companies House, company 05956777

Latka — [getlatka.com/companies/justpark](https://getlatka.com/companies/justpark)

SpotHero business model — [vizologi.com/business-strategy-canvas/spothero-business-model-canvas](https://vizologi.com/business-strategy-canvas/spothero-business-model-canvas)

### India

Inc42 — [inc42.com/company/park](https://inc42.com/company/park)

Get My Parking — [getmyparking.com/about.php](https://getmyparking.com/about.php)

Parkobot launch — [startupsuccessstories.in](https://startupsuccessstories.in)

### Operations & IoT

Parkable access control — [parkable.com/nz/parking-management-system/features/access-control](https://parkable.com/nz/parking-management-system/features/access-control)

BookFlowGo integration — [parking.net/parking-news/bookflowgo/what-integration-really-takes](https://parking.net/parking-news/bookflowgo/what-integration-really-takes)

Ronspot ANPR integration — [ronspotflexwork.com/blog/anpr-parking-integration](https://ronspotflexwork.com/blog/anpr-parking-integration)

Airbnb smart lock integrations — [news.airbnb.com](https://news.airbnb.com)

Hubble — device provisioning at scale

Memfault — OTA testing without bricking devices